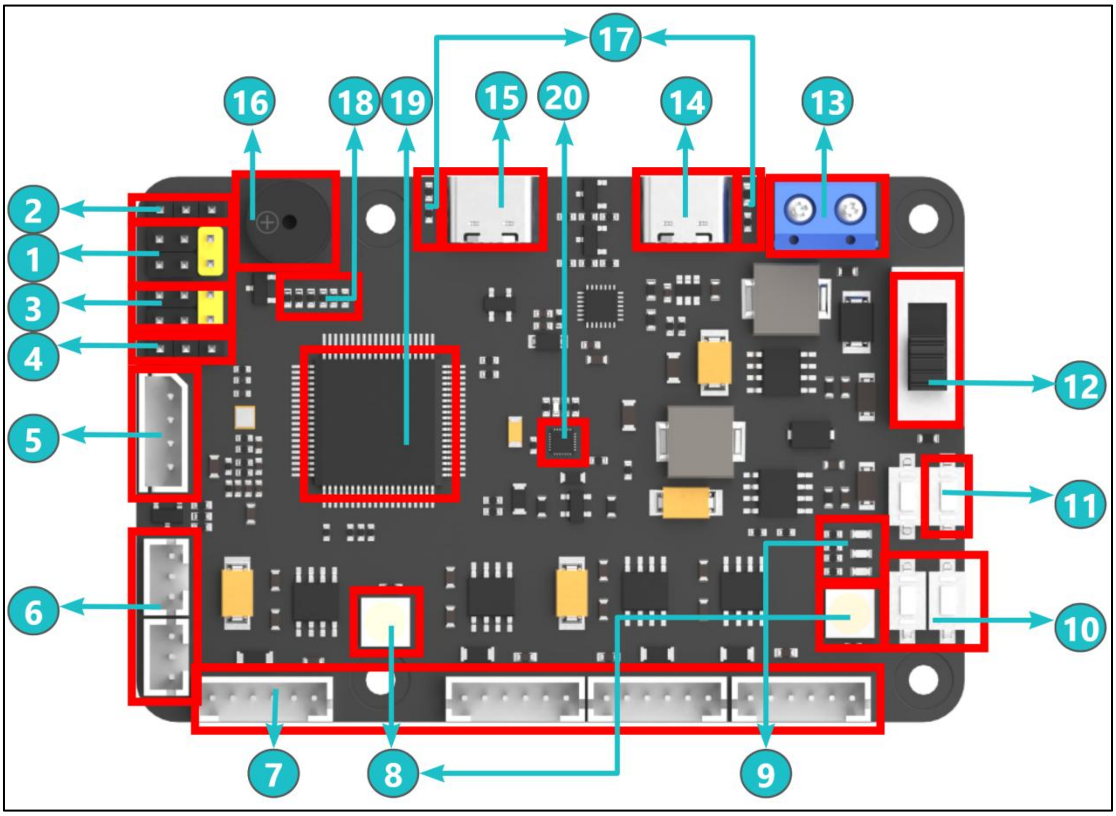
RRCLite Hardware Introduction

1. Development Board Diagram



NO.	Function Description
(1)	First group of PWM servo interface x2: Control high-current and high-torque servos
(2)	Power selection pin for the first group of PWM servos: By shorting with a jumper cap, the power supply for the servos can be selected between the power supply voltage and 5V
(3)	Second group of PWM servo interface x2: Control high-current and high-torque servos
(4)	Power selection pin for the second group of PWM servos: By shorting with a jumper cap, the power supply for the servos can be selected between the power supply voltage and 5V
(5)	I2C expansion interface: Used for communication with I2C interface modules

(6)	Bus servo interface x2: Control the bus servos to move the robotic arm
(7)	4-Ch encoder motor interface x4: Drive four motors simultaneously. The connection method varies depending on the robot car type. For specific connection instruction, please refer to the corresponding tutorials.
(8)	RGB light x2: The color of the RGB light can be programmed.
(9)	User Indicator: Custom LED light functions
(10)	User button x2: Custom button functions and combine button functions
(11)	Reset button: Reset button for the controller chip
(12)	Power switch: Power switch for the development board
(13)	Power interface: DC 5V ~ 12.6V power input; can be powered by power supply or battery
(14)	5V5A external power supply: Specially designed to power Raspberry Pi, Jetson Nano development boards, etc.
(15)	USB serial port 1/flashing and downloading: Used for serial port flashing and communication
(16)	Buzzer: Used for user prompts and alarm functions.
(17)	On-board power indicator: Indicates whether the voltage of each part is normal
(18)	GPIO expansion and SWD debugging: Used for other external module expansion and debugging
(19)	STM32F407VET6 controller chip: 168MHz main frequency, ARM Cortex-M4 core, 512KB Flash, 192KB RAM, and 82 common IO ports
(20)	QMI8658 IMU attitude sensor: Provides the current attitude of the development board

2. FAQ for Controller

Q1: How to control the control board with Raspberry Pi or Jetson Nano? How to communicate with the control board?

A: Raspberry Pi, Jetson Nano, and similar devices can communicate with the control board via the serial port 1.

Please note that the serial port 2 is a 5V5A power output port dedicated to Raspberry Pi and other development boards. It cannot used for serial port communication.

Q2: Which functions on the development board are managed by the microcontroller?

A: The microcontroller on the development board manages several components, including the buzzer, robotic arm, servo pan-tilt, buttons, encoder motor interface, QMI8658 accelerometer and gyroscope, USB serial port, display interface, I2C interface, RGB LED, LED, GPIO expansion port, and more.

Q3: How to learn to use the development board? How to update the firmware of the development board?

A: The development board comes with rich comprehensive tutorials, such as FREERTOS system LED control tutorial, encoder motor control tutorial, and motor PID control tutorial. The interfaces have been implemented, which can achieve rapid development. To update the firmware, you can use the USB serial port for downloads and JLink for debug updates. All interfaces are externally connected to make learning and debugging easier.